F. No. 318/84/2024- GCRT Part (3)
Government of India
Ministry of New & Renewable Energy

Atal Akshay Urja Bhawan
Lodhi Road, New Delhi-110003
Date: 6th May 2026

OFFICE MEMORANDUM

Subject: - Standard Operating Procedure (SOP) for Warranty Management of Solar PV Modules installed under Distributed Renewable Energy Schemes of the MNRE - reg.

The (MNRE) has been implementing several Distributed Renewable Energy (DRE) Schemes across the country such as PM Surya Ghar: Muft Bijli Yojana (PMSG: MBY), PM KUSUM, New Solar Power Scheme, etc. Several representations have been received in the Ministry regarding standard procedure for warranty on solar PV modules as manufacturers have different procedures and criteria for discharging warranty claims on solar PV Modules manufactured by them.

2 In this regard, an SOP for Warranty Management and Claims Settlement of Solar PV Modules being installed under various DRE Schemes of the MNRE has been prepared and the same is attached at Annexure. This SOP shall be applicable to various DRE schemes of the MNRE such as PMSG: MBY, PM KUSUM, New Solar Power Scheme, etc. All solar PV module manufacturers shall adhere to the provisions of this SOP for warranty management and claims settlement for the solar PV modules manufactured by them and are being installed under these schemes of MNRE.

3 This issues with approval of competent authority.

Anubhav
(Anubhav Uppal)
Scientist-D

To

All manufactures of solar PV modules

Copy to

i. PPS to Mission Director, PMSG: MBY
ii. ED (PMSG: MBY), REC ltd
iii. DG, NISE with a request to implement necessary provisions in the DCR portal and ensure effective implementation of this SOP
iv. Manufacturing Division, MNRE with a request to kindly circulate the SOP with all Solar PV manufacturers.

Annexure

Standard Operating Procedure (SOP) for Warranty Management and Claims Settlement of Solar PV Modules installed under Distributed Renewable Energy Schemes of the Ministry of New and Renewable Energy (MNRE)

1. **Applicability**

This Standard Operating Procedure (SOP) shall be applicable to various distributed renewable energy schemes of the MNRE such as PM Surya Ghar: Muft Bijli Yojana, etc. All solar PV module manufacturers participating directly in these schemes, or supplying to such entities who explicitly give in writing to the solar PV manufacturer, that the solar PV module being procured is intended for deployment in MNRE's distributed renewable energy schemes, shall adhere to the provisions of this SOP for warranty management and claims settlement.

2. **Limited Product Warranty**

Subject to the applicable terms, conditions, and limitations, the manufacturer shall warrant that the solar PV modules are free from defects in design, materials, workmanship, or manufacturing that materially affect their functionality, and that the modules conform to the applicable technical specifications and approved drawings, for a minimum standardized warranty period of ten (10) years from the Warranty Commencement Date. The Warranty Commencement Date shall be the date of installation or commissioning of the solar PV system, subject to the condition that the solar PV module after being sold by the solar PV manufacturer, is deployed on the project site within six months from the date of invoice of sale by the solar PV manufacturer; in case of this period being more than six months, the warranty period of ten years shall stand reduced commensurately.

3. **Testing and Commissioning Protocol**

To ensure early detection of manufacturing and installation-related defects and to minimize future warranty disputes, standardized testing shall be conducted by the vendors/manufacturers at the time of installation/commissioning of rooftop solar PV systems or at time of warranty claims.

a. **Testing of Solar PV Modules**:

- **Electroluminescence (EL) Test**: EL test in the field is commonly performed on solar PV modules for detection of micro-cracks in solar cells, broken cells, soldering defects, finger interruptions, inactive or weak cells etc. if any. If any such defect is found out, then that panel may not to be used.
- **I-V Test**: an I-V/Current-Voltage test is commonly performed in the field on solar PV module to verify performance and compare actual output with expected values. For any warranty claims, field level I-V Tests on PV modules may be performed to check the wattage. Warranty claim will be applicable in case:

  Measured wattage < (Nameplate wattage – manufacturer specified tolerance – applicable degradation – measurement uncertainty)

- Any other relevant test.

b. **Testing Standards**:

- All testing procedures shall strictly comply with applicable BIS/ IEC standards.
- Testing methodology, measurement procedures, and acceptance criteria shall be aligned with applicable IS/IEC module performance standards and ALMM technical requirements.

c. **Testing Agencies and Laboratory Requirements:**

- Wherever laboratory validation is required, testing shall be conducted only through National Accreditation Board for Testing and Calibration Laboratories (NABL) empanelled laboratories.

4. **Limited Power Output Warranty**

Subject to the applicable terms, conditions, and limitations, the manufacturer shall warrant that, for a period of twenty-five (25) years from the Warranty Commencement Date, the solar PV modules shall maintain the specified minimum power output performance, with degradation limits measured under Standard Test Conditions (STC). In case, the Warranty commencement date is after six months from the date of invoice of sale by the solar PV manufacturer, the period of 25 years, the warranty period of 25 years shall stand reduced commensurately proportional to such period in excess of six months from invoice by solar PV module manufacturer. The performance warranty, degradation trajectory, testing methodology, and acceptance criteria shall be aligned with applicable Indian Standards (IS), IEC standards on module performance and degradation, and the technical compliance requirements prescribed under the Approved List of Models and Manufacturers (ALMM), in order to ensure uniformity, technical consistency, and comparable warranty obligations across manufacturers.

5. **Warranty Exclusions**

a. **Critical Warranty Exclusion Factors**

- Non-compliance with the manufacturer's installation manual, applicable IEC/IS standards, or the use of sub-standard mounting structures may constitute a warranty exclusion only where such non-compliance is demonstrably established against measurable technical parameters prescribed under relevant IEC/IS standards. Rejection of warranty claims on these grounds shall be permitted only where the manufacturer provides documented technical evidence establishing a clear and direct link between the identified installation deficiency or structural inadequacy and the reported module failure or performance degradation.
- Forbidden repairs, modifications, and component replacement.
- Installation in mobile devices (excluding approved PV tracking systems), marine environments, or for non-standard applications shall constitute a warranty exclusion unless such usage is expressly permitted under the applicable technical specifications. For the purpose of warranty exclusion, "marine environment" shall be clearly defined by the manufacturer based on measurable technical criteria, including proximity to coastline, salinity exposure levels, corrosion classification, or other relevant parameters, so as to avoid ambiguity for coastal and near-coastal installations.
- Force Majeure events shall include only natural disasters (such as

earthquakes, floods, cyclones, etc.). Supply chain disruptions, including those arising due to wars, riots, or related commercial and logistical impacts, shall not be treated as Force Majeure for the purpose of warranty exclusion.

- Exposure to improper voltage, power surges, where such conditions are established through measurable and test-based criteria aligned with applicable IS/IEC standards, shall be treated as warranty exclusion factors. Exposure to abnormal environmental conditions, where such conditions are established through measurable and test-based criteria aligned with applicable IS/IEC standards, shall be treated as warranty exclusion factors only for installations located in technically classified extreme industrial corrosion zones supported by documented technical justification.
- Tempering of nameplate, module serial number on Solar PV modules.
- Glass breakage resulting from verified external or man-made damage, supported by demonstrable evidence, shall be treated as a warranty exclusion.
- Failure to match the shipping address as per the invoices
- Major warranty exclusions include factors such as misuse, neglect, tampering, or improper handling.

6. Complaint Handling Mechanism

a. Complaint Registration and Tracking

- Complaints shall be formally logged through email or any other online portal of the manufacturer.
- Solar PV module manufacturer shall provide customer care/helpline number (phone/WhatsApp/e-mail) to address consumer complaints.
- All manufacturers shall create an online portal wherein the consumer can register. More than one solar module manufacturers can also come together and create one such common portal.
- The manufacturer shall ensure ease of availability of consumer level Warranty Certificate for the preferably through digital mode
- All manufacturers shall create an online portal where in the consumer can register for warranty of individual solar PV modules installed at its site. Also, consumer shall be able to download individual module wise warranty certificates from this portal.
- The B2C DCR Certificate available on the DCR Portal shall serve as a DCR-cum-Warranty Certificate. The approved SOP shall be incorporated as part of the existing DCR Certificate.
- The manufacturer portal for warranty certificate generation may be integrated with the DCR portal—either through API-based integration or by enabling manufacturers to download relevant data through their login on the DCR portal and upload the same on their own portal—to fetch data pertaining to already generated B2C certificates. Such data may include consumer name, DCR certificate number, panel details, mobile number, and vendor details.

- The mobile number of the consumer, as entered in the DCR portal at the time of B2C certificate generation, may be utilized for consumer login (through OTP authentication) on the manufacturer's portal. This will obviate the need for separate registration or repeated visits by the consumer for warranty certificate generation
- For manufacturers without a dedicated warranty portal, a provision may be enabled in their login on the DCR portal to access and download relevant B2C certificate data (including consumer details, certificate number, panel number, mobile number, and vendor details). This data may be stored by the manufacturers and subsequently utilized upon development of their own portal. Such data shall also serve as a primary source of authentication for processing warranty claims
- The generated DCR-cum-Warranty Certificate may also be made available in the consumer login on the National Portal which can be fetched from DCR portal at the time of installation while fetching panel-level details.
- Manufacturers shall be allowed to submit their own warranty certificate format, aligned with this SOP. A provision shall be made on the DCR portal (under Manufacturer's login) to upload such format.
- The uploaded formats shall be automatically appended to B2C certificate corresponding to that manufacturer's panels.
- In cases where manufacturers do not provide a separate warranty certificate format, this SOP shall be attached by default as the warranty certificate.
- All complaints shall be registered in the manufacturer's online portal with a unique reference number.
- The concerned department of the solar PV module manufacturer shall register complaints immediately upon receipt through email or any approved online portal.
- Complaint acknowledgement shall be provided to the customer within a standard response timeline of 24 hours to 48 hours.
- Response timelines shall be communicated to the customer at the time of acknowledgment.
- Fixed timelines to address consumer complaints may be made available periodically on online portal i.e. 24 to 48 hours for response, 7days for inspection by the manufacturer, 30 days for replacement if required, etc. further, an escalation matrix may be clearly defined by the manufacturer.
- Customers shall provide a valid invoice and proof of purchase.
- DCR/B2C certificate may also be considered equivalent to invoice in case of loss to the original Invoice/Bill.
- Module serial number shall be intact and legible.
- Installation/commissioning report shall be submitted.
- High-quality photographs of the defective PV module shall be provided.
- For low power output complaints, details of the PV generator, inverter, electrical circuitry/layout, and site shadow conditions shall be submitted.

- Standard reporting templates shall be maintained by the manufacturer to ensure traceability and proper record keeping.
- Automatic closure of complaints due to non-response by consumer shall only be undertaken in accordance with prescribed communication procedures and documented attempts at contact.

b. **Quality Assurance, Technical Investigation and Testing**

- The QA/QC (Quality Assurance / Quality Control) team shall lead technical investigation, root cause analysis (RCA), and preparation of Corrective and Preventive Action (CAPA) reports, with clearly defined timelines for each stage of the process.
- Technical assessment and RCA shall be completed within a standardized timeline of 7–10 working days, wherever feasible.
- A standardized CAPA template shall be adopted to ensure uniformity and consistency in investigation and reporting.
- The QA/QC team shall coordinate site inspections and third-party testing, wherever required for verification of defects or performance issues.
- Third-party testing, where required, shall be conducted through approved laboratories such as NABL accredited laboratories, as notified or recognized under applicable standards.
- Clear escalation and dispute resolution mechanisms shall be followed in cases requiring independent technical verification.
- Regarding third-party testing costs; manufacturers and consumers may initially bear testing costs in the ratio of 50:50, with ultimately full testing cost being born by the manufacturer if the complaint w.r.t. solar PV module, as claimed by the consumer, is conclusively established as correct after testing and full testing cost being born by the consumer, if the complaint w.r.t. solar PV module, as claimed by the consumer, is conclusively established as incorrect/fraudulent or technically untenable. Before proceeding for testing, the manufacturer and the consumer will sign a contract, wherein both the parties will mutually agree that it will be legally mandatory for the party found at fault after the result of the testing, that it shall bear the full cost towards testing.
- The QA/QC team shall maintain a complaint investigation register along with complete technical documentation and records for traceability.

c. **Claim Evaluation & Resolution**

- The concerned team/department shall review the CAPA report, investigation findings, and site observations to undertake evaluation of the warranty claim.
- Site visits shall be conducted where visual inspection is insufficient.
- Acceptance or rejection of warranty claims shall be finalized based on defined and uniform evaluation criteria.
- The outcome of the evaluation shall be formally communicated to the

consumer along with supporting findings and justification.

- In case the warranty claim is accepted, the concerned team/department shall initiate further warranty processing and replacement/repair.
- Where the consumer disputes the conclusion, additional technical engagement, including reassessment or site inspection, may be undertaken as required.

d. **Remedies & Replacement**

- Repair or replacement of defective modules shall be completed within 30 days
- Where replacement requires fresh production, supply shall be completed within 90 days, in accordance with agreed contractual terms.
- In case of module replacement during the warranty period, manufacturers shall ensure supply of modules with equivalent or higher technical specifications, compatible size, and electrical characteristics so as to maintain system performance, safety, and design compatibility. In case such solar PV module(s) is/are not being manufactured by the manufacturer at the relevant time, the manufacturer can compensate the consumer by paying the commensurate cost to the consumer.
- Replacement modules shall be new modules only and shall not be refurbished, repaired, or re-manufactured units. The replaced module shall continue to be covered under warranty as per the original warranty specifications and remaining warranty period.
- Refund of the product cost may be provided in cases where repair or replacement is not feasible. In case of irreparable manufacturing defects, refund shall be made at the original purchase price without application of annual depreciation.
- Supply of additional modules may be undertaken, where technically appropriate, to compensate for verified power loss.
- For verified manufacturing defects, manufacturers shall bear reasonable logistics and transportation costs associated with replacement. Labour and ancillary cost norms related to removal, installation, insurance, and handling.
- Warranty applicability shall remain valid where the module continues to be installed at the original project location.
- Warranty transfer shall be permitted where legal documentation supports ownership transfer and module traceability is maintained.

7. **Limited Liability**

To the maximum extent permitted by applicable law, the Manufacturer shall not be liable for any damage or injury to persons or property, or for any indirect, incidental, or consequential losses arising from or related to the use, installation, or performance of the solar module, including any defects in the module.

8. **Force Majeure**

The Manufacturer shall not be liable for any failure or delay in performance under this Warranty caused by force majeure events (such as earthquakes, floods, cyclones, etc.). Supply chain disruptions, including those arising due to wars, riots,

or related commercial and logistical impacts, shall not be treated as Force Majeure for the purpose of warranty exclusion.

9. **Continuity of Warranty Obligations**

Manufacturers shall ensure continuity of warranty commitments throughout the warranty tenure, including the 25-year performance warranty period. In the event of company restructuring, merger, insolvency, or dissolution, appropriate legal or financial mechanisms shall be established to honour warranty obligations.

10. **Institutional Liability**

DISCOMs, or implementing agencies shall not be party to warranty claims, disputes, or settlement processes. Warranty redressal shall remain a direct contractual and operational responsibility between the consumer and the manufacturer/vendor as per the contractual/ warranty related, agreements/documents, executed amongst these parties.

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